BRANDON SANTOR

Geospatial researcher | Software developer | U.S. Marine Corps veteran

emp1241.github.io

Builds scientific tools from the data pipeline to the interface: tree-ring and climate workflows, deep-ocean WebGIS, reconstructed Martian terrain, spatial evidence systems, and independent GNSS research. Combines research leadership with application architecture, source processing, and reproducible handoffs.

RESEARCH & SOFTWARE EXPERIENCE

Research Assistant & Software Developer | Torbenson Lab, Texas A&M

March 2026-present | Paid appointment since June 2026

- Independently created Tree-Ring Atlas Explorer (TRAE) in response to lab research needs; the prototype led to a paid appointment for continued development and research support.
- Built reviewed scientific imports, linked maps and graphs, climate matching, provenance, sessions, and exports. Added PRISM, NetCDF, ROXAS-output preparation, and WalDendro cleaning workflows.
- Used TRAE to derive, clean, organize, and export data for a co-authored manuscript in preparation.

Research Lead & GIS Developer | NWSW, Texas A&M GIS Practicum

Fall 2026-present

- Lead cross-team synthesis, source reconciliation, reusable boundary preparation, geographic joins, and evidence-matrix integration.
- Used my Research Architect framework to generate the coordination system, with versioned context, SHA-256 checks, and reviewed contributor handoffs.
- Independently built a separate FastAPI/PostGIS/TypeScript/Leaflet evidence application for source inspection, spatial investigation, and context-preserving export.

Mars Explorer | Team Lead & Database / 3D Systems Developer

Curiosity / Gale crater | Local prototype

- Lead a team of five; design and implement the 41-table spatial database, source-processing pipelines, read-only API, and interactive 3D terrain system.
- Transform orbital elevation and imagery into textured meshes, normals, UV coordinates, and movement-height data using Python, NumPy, Rasterio/GDAL, and custom Cesium geometry.

EDUCATION

Texas A&M University | B.S. Geographic Information Science & Technology

Expected May 2027 | Institutional GPA: 4.0 / 4.0

A.S. Computer Networking | ITT Technical Institute | 2007

SELECTED WORK & EXPERIENCE

Research software, field analysis, technical instruction, and operational leadership

INDEPENDENT SOFTWARE & RESEARCH

Thermal Vent Atlas Explorer | Independent Creator & Developer

Sylvan Lab, Texas A&M Oceanography

- Delivered an online/offline research WebGIS connecting 742 atlas records with bathymetry, tectonic boundaries, geomicrobiology context, comparisons, sessions, and scoped exports.

COLE | Project Lead, Application Design & Feature Integration

Cost Of Living Explorer | Functioning local application

- Lead household-driven county search and feature add-on integration, combining public costs, climate, geographic lookup, and hazard context with visible assumptions and three-county comparison.

PhalAInx | Independent Software Studio

Creator & Principal Designer

- Created Research Architect: research-system generation, scientific method modules, context continuity, and reviewed handoffs, informed by a technical white paper and comparative literature review.
- Created High Command, my multi-LLM desktop collaboration product, now in active testing.

The Deception Gap | Independent Continued GNSS Research

Expanded class project | 781 retained readings, three smartphones, five crowd categories

- Led continued field collection, statistical analysis, interpretation, and poster development; compared actual horizontal error with device-reported uncertainty.

PROFESSIONAL & MILITARY EXPERIENCE

Independent Esri GIS Technical Consultant & Instructor

February-May 2026

- Audited seven graduate GIS modules; rebuilt six workflows into 189 pages of illustrated manuals. Corrected projections, raster alignment, schemas, joins, licensing, and desktop-to-web dependencies.

U.S. Marine Corps | Infantryman, Acting Platoon Sergeant & Logistics NCO

2011-2013

- Managed logistical support, inventory, and resource allocation for approximately 350 Marines; coordinated personnel and field operations under strict accountability and safety requirements.

TEKsystems | Network & Computer Systems Contractor | 2007-2011
Enterprise hardware/software deployments involving 100+ devices per project.

TECHNICAL TOOLKIT & ADDITIONAL LEADERSHIP

Python, pandas, NumPy, Rasterio; JavaScript/TypeScript; PostgreSQL/PostGIS; FastAPI/Flask; Cesium/Leaflet; ArcGIS Pro/Field Maps; QGIS/GDAL; Electron/React/SQLite; scientific import/export; research provenance.

NASA SUITS proposal (2021-2022): Team lead and primary system designer for a conceptual HoloLens 2 EVA interface; co-authored technical documentation submitted to NASA. **TXGIS Day (2025):** Second Place for a Health Geography GIS/statistical analysis presentation.